

When to Alert, When to Defer: Conformal Selective Triage for ICU Mortality in Sepsis-Associated AKI

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Abstract

Objective: To develop and evaluate an uncertainty-aware triage framework for early in-hospital mortality prediction in sepsis-associated acute kidney injury (SA-AKI), with primary emphasis on safe automation boundaries rather than raw discrimination alone.

Materials and Methods: We analyzed 10036 ICU-stay-level SA-AKI cohort records from `mimic_saaki_raw.v2.csv` scored at 24 hours after ICU admission, with event prevalence 0.267. All deployment claims used subject-grouped train/validation/test splits by `subject_id`. Conventional baselines included logistic regression, XGBoost, a grouped-Optuna-tuned LightGBM benchmark (40 trials), an exploratory CatBoost diversity comparator, and score-only baselines. Calibration was selected on grouped validation data. We evaluated disagreement-based selective triage as an ablation and Mondrian conformal selective triage as the main method. We summarized coverage, alert positive predictive value (PPV), clear negative predictive value (NPV), calibration, decision-curve utility, subgroup heterogeneity, and robustness under simulated distribution shift, with headline conformal estimates repeated across 21 grouped seeds.

Results: The best conventional discriminative baseline was XGBoost with isotonic calibration (mean AUROC 0.768; recall at $PPV \geq 0.50$ of 0.608). Grouped Optuna improved LightGBM validation recall at $PPV \geq 0.50$ to 0.590, but repeated grouped benchmarking still preserved only a narrow tree-model performance band. The prediction ceiling remained limited: even when `time_to_event_hrs` was added as a leakage-like feature, AUROC reached only 0.810. Single-model conformal triage achieved coverage 0.952 at $\alpha = 0.05$ while retaining a clinically meaningful defer region. The multi-model union consensus at $\alpha = 0.05$ concentrated decisions into a smaller, more reliable actionable region with alert PPV 0.729 and clear NPV 0.962. At a decision threshold of 0.20, the selected continuous model achieved net benefit 0.133 versus 0.082 for APACHE-III and 0.082 for SOFA.

Discussion and Conclusion: The main contribution is not a new AUROC leader; it is a deployment-oriented Alert/Defer/Clear framework that quantifies when the model should defer. Conformal selective triage is a stronger primary story than disagreement-based deferral because it supplies finite-sample coverage guarantees and more graceful degradation under shift.

Keywords conformal prediction, sepsis-associated acute kidney injury, mortality prediction, intensive care, uncertainty quantification, deployment

Abbreviations AKI, APACHE-III, AUROC, ICU, MIMIC, NPV, PPV, SA-AKI, SOFA

Key Messages

- First-24-hour SA-AKI mortality discrimination is bounded even after grouped Optuna tuning and an added CatBoost benchmark.
- Conformal selective triage reframes ICU risk prediction into Alert/Defer/Clear actions with coverage guarantees.
- Conservative multi-model consensus and explicit deferral remain more reliable under simulated shift than fixed-threshold policies.

Introduction

Predicting mortality in critically ill patients with sepsis-associated acute kidney injury is clinically important, but retrospective machine-learning studies often stop at aggregate discrimination and overstate deployment readiness. In practice, clinicians need to know not only how well a model ranks patients on average, but also when it should escalate care, when it can safely clear a patient from immediate concern, and when it should defer because the case is too uncertain for automation. The working cohort in this study is a MIMIC-derived

SA-AKI dataset with 10036 T24 rows, outcome prevalence 0.267, and subject-level duplication that makes grouped evaluation mandatory.

This project starts from a hard empirical observation: discrimination in this cohort is bounded. The strongest conventional models occupy a narrow performance band, with XGBoost at AUROC 0.768, LightGBM at 0.765, and CatBoost at 0.768. Even a leak-prone ceiling experiment that includes `time_to_event_hrs` reaches only AUROC 0.810, compared with 0.763 in the honest grouped setting. That finding changes the research objective. The key question is no longer

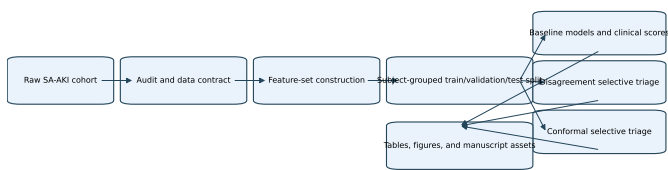


Figure 1 Cohort definition and evaluation workflow.

how to squeeze another 0.01 of AUROC from the same first-24-hour features; it is how to build a clinically legible triage policy that behaves safely under bounded signal.

Two uncertainty-aware strategies are therefore compared. The first is a disagreement-based selective-triage baseline that treats divergence between a physiology-only model and a process-enriched model as a signal to defer. The second, and primary contribution of this paper, is conformal selective triage. Conformal prediction converts probabilistic risk scores into set-valued decisions with finite-sample coverage guarantees, which naturally supports a three-way clinical interface: Alert, Clear, or Defer.,

The manuscript makes three contributions. First, it documents the prediction ceiling for first-24-hour SA-AKI mortality risk in a subject-grouped evaluation design. Second, it introduces conformal selective triage as the main decision framework and shows that it provides explicit safety-oriented operating points in terms of coverage, alert PPV, and clear NPV. Third, it demonstrates that multi-model conformal consensus and conformal deferral degrade more gracefully than fixed thresholds under missingness and process-feature perturbations.

Methods

Study Design and Data Source

This retrospective study used `mimic.saaki_raw.v2.csv`, a T24 SA-AKI cohort derived from MIMIC. One row represents one ICU-stay-level SA-AKI record scored at 24 hours after ICU admission. The primary label is in-hospital mortality (`event_observed`), and the time-to-event variable is `time_to_event_hrs`. Because repeated patients remain in the cohort, all deployment claims use subject-grouped evaluation by `subject_id`.

Cohort and Evaluation Workflow

The analysis pipeline began with an audit and data-contract pass, followed by feature-set construction, subject-grouped train/validation/test splitting, conventional benchmark modeling, disagreement-based selective triage, conformal selective triage, and manuscript-asset generation.

Baselines

Conventional baselines included logistic regression, LightGBM, and XGBoost. Calibration was selected on grouped validation data by benchmarking sigmoid and isotonic recalibration on a held-out grouped validation split. We also retained score-only comparators based on SOFA and APACHE-III where those variables were available, treating them as single-feature logistic baselines under the same grouped protocol.,

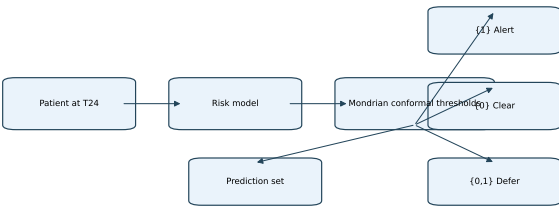


Figure 2 Conformal selective-triage decision flow.

To close the remaining methodological rigor gap from earlier drafts, we added a grouped Optuna search for the main LightGBM benchmark. The search used 40 trials and optimized a deployment-oriented objective that combined AUROC, recall at $PPV \geq 0.50$, and Brier score on grouped validation data. The tuned LightGBM configuration was then frozen and carried into the repeated grouped benchmark. We also added CatBoost as a benchmark-only diversity comparator to test whether a distinct gradient-boosting family materially changed the model-selection story.

The disagreement-based selective-triage comparator follows the earlier project draft. A process-enriched full model and a physiology-severity model were trained on the same grouped split. Cases were considered actionable only when the two models agreed closely enough on grouped validation data.

Conformal Selective Triage

The main method is Mondrian conformal prediction., A training subset fits the risk model, a separate calibration subset estimates nonconformity thresholds, and the test subset receives set-valued predictions. For binary mortality prediction, the prediction set can be one of three clinically interpretable outcomes:

- {1}: Alert
- {0}: Clear
- {0,1}: Defer

Table 2 Baseline discrimination and calibration benchmark.

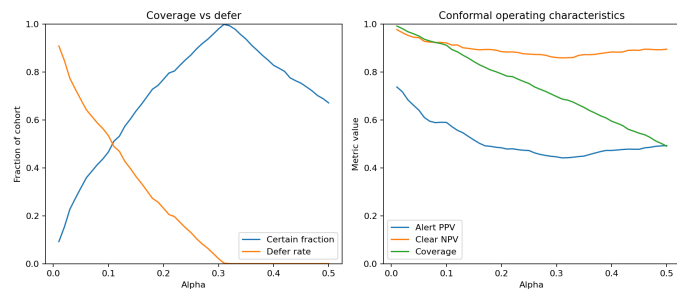
Model	AUROC	AUPRC	Brier	ECE	Calibration slope	Recall @ PPV _L =0.50
apache_iii_score	0.579	0.330	0.245	0.226	0.843	0.023
lightgbm	0.765	0.545	0.161	0.021	1.031	0.575
logistic	0.751	0.532	0.165	0.026	1.140	0.556
sofa_total_24hr	0.646	0.397	0.233	0.217	1.017	0.146
xgboost	0.768	0.544	0.161	0.017	1.035	0.608

Table 3 Main conformal results across repeated grouped splits.

alpha	Seeds	Coverage mean	Coverage 2.5%	Coverage 97.5%	Certain fraction	Alert PPV	Clear NPV	Miss count
0.050	21.000	0.952	0.945	0.959	0.318	0.643	0.948	23.095
0.100	21.000	0.900	0.882	0.914	0.503	0.579	0.924	49.952
0.150	21.000	0.848	0.826	0.871	0.653	0.540	0.903	80.190
0.200	21.000	0.797	0.771	0.821	0.782	0.509	0.887	109.000

Table 4 Multi-model consensus conformal results.

Ensemble	Base model(s)	alpha	Seeds	Coverage mean	Certain fraction	Alert PPV	Clear NPV	Miss count
intersection	lightgbm+xgboost+logistic	0.050	21	0.911	0.462	0.591	0.926	45.048
intersection	lightgbm+xgboost+logistic	0.100	21	0.835	0.683	0.534	0.901	86.000
intersection	lightgbm+xgboost+logistic	0.150	21	0.766	0.820	0.501	0.885	116.762
intersection	lightgbm+xgboost+logistic	0.200	21	0.699	0.888	0.486	0.876	134.000
single_model	xgboost	0.050	21	0.952	0.318	0.643	0.948	23.095
single_model	xgboost	0.100	21	0.900	0.503	0.579	0.924	49.952
single_model	xgboost	0.150	21	0.848	0.653	0.540	0.903	80.190
single_model	xgboost	0.200	21	0.797	0.782	0.509	0.887	109.000
union	lightgbm+xgboost+logistic	0.050	21	0.982	0.164	0.729	0.962	8.714
union	lightgbm+xgboost+logistic	0.100	21	0.954	0.308	0.651	0.949	21.381
union	lightgbm+xgboost+logistic	0.150	21	0.922	0.432	0.606	0.933	38.190
union	lightgbm+xgboost+logistic	0.200	21	0.885	0.550	0.568	0.918	58.286

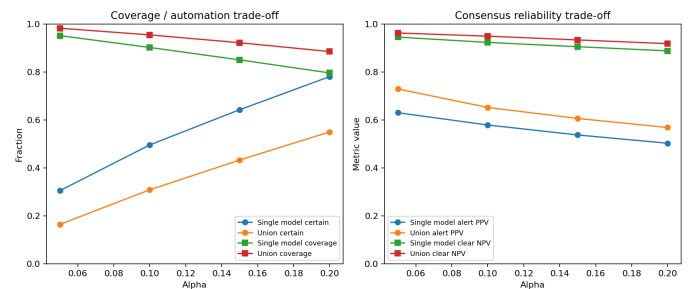
**Figure 6** Conformal operating-characteristic curve.

At $\alpha = 0.05$, single-model conformal triage achieved coverage 0.952 with certain-decision fraction 0.318, alert PPV 0.643, and clear NPV 0.948. Under the manuscript sweet-spot criterion ($\text{NPV} \geq 0.90$ and $\text{PPV} \geq 0.55$), the best operating point occurs at $\alpha = 0.120$ with certain-decision fraction 0.532, alert PPV 0.556, and clear NPV 0.912.

Multi-Model Consensus

Table 4 and Figure 7 show the consensus extension. The union of conformal prediction sets is deliberately conservative: it acts on fewer patients but produces a more reliable actionable region.

At $\alpha = 0.05$, the union consensus yields coverage 0.982, alert PPV 0.729, and clear NPV 0.962, confirming that consensus amplifies reliability at the cost of automation rate.

**Figure 7** Multi-model consensus trade-off panel.

Comparison With Disagreement-Based Selective Triage

The disagreement-based policy remains an important ablation because it shows that explicit deferral helps, but its guarantees are empirical and heuristic rather than finite-sample. The repeated grouped summary is included in Appendix B. In brief, disagreement triage narrows the actionable region and reduces actionable error, but conformal triage supplies the stronger methodological framing because it binds that defer region to a formal coverage objective.

Robustness Under Distribution Shift

Table 5 and Figure 8 summarize robustness under simulated shift.

As extra missingness is injected, conformal coverage remains stable because the framework defers more cases. By contrast, fixed thresholds

Table 5 Shift robustness versus fixed-threshold policies.

Scenario	Severity	Conformal coverage	Conformal certain fraction	Conformal alert PPV	Conformal clear NPV	Fixed PPV @0.50	Fixed recall @0.50
clean	0.000	0.950	0.316	0.640	0.943	0.593	0.294
physiology_missingness_shift	0.100	0.950	0.297	0.597	0.935	0.573	0.252
physiology_missingness_shift	0.200	0.963	0.268	0.632	0.938	0.618	0.207
physiology_missingness_shift	0.300	0.970	0.235	0.663	0.925	0.619	0.171
measurement_process_dropout	1.000	0.950	0.281	0.646	0.943	0.596	0.309
care_process_dropout	1.000	0.947	0.320	0.624	0.945	0.606	0.301

Table 6 Subgroup heterogeneity summary for conformal triage at $\alpha = 0.10$.

Subgroup	Level	n	Event rate	Coverage	Certain fraction	Alert PPV	Clear NPV
age_band	51-65	556	0.257	0.903	0.523	0.571	0.902
age_band	66-80	776	0.277	0.912	0.448	0.603	0.926
age_band	i=50	244	0.201	0.930	0.602	0.621	0.949
age_band	i=80	425	0.292	0.908	0.344	0.571	0.913
gender	F	954	0.257	0.921	0.454	0.620	0.928
gender	M	1047	0.273	0.902	0.477	0.564	0.914
mech_vent	-1	916	0.271	0.904	0.457	0.558	0.928
mech_vent	1	1085	0.261	0.917	0.473	0.620	0.915
sofa_band	High SOFA	566	0.412	0.866	0.484	0.659	0.842
sofa_band	Low SOFA	855	0.171	0.949	0.480	0.469	0.929
sofa_band	Medium SOFA	580	0.262	0.900	0.428	0.505	0.952
vasopressor	-1	1277	0.230	0.926	0.486	0.586	0.932
vasopressor	1	724	0.327	0.885	0.431	0.593	0.887

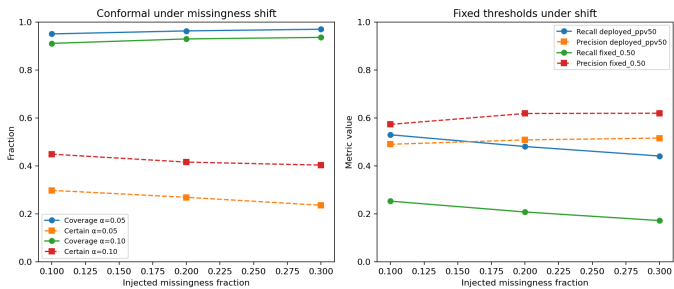


Figure 8 Robustness-under-shift panel.

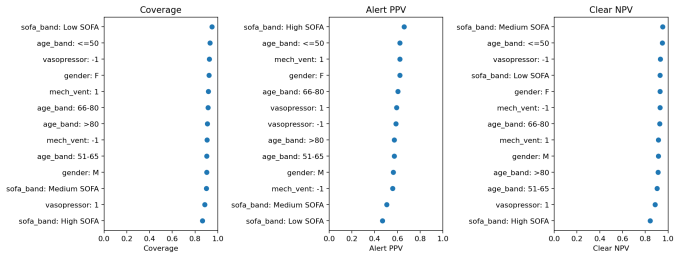


Figure 9 Subgroup coverage and reliability forest plot.

lose recall while offering no explicit warning that uncertainty has increased.

Subgroup Heterogeneity

Table 6 and Figure 9 summarize subgroup heterogeneity for conformal triage at $\alpha = 0.100$.

Coverage and clear NPV remain high across most demographic subgroups, but high-acuity strata are predictably harder. Reporting subgroup summaries at $\alpha = 0.100$ exposes that trade-off explicitly:

low-SOFA strata preserve high coverage with a reasonable action rate, whereas high-SOFA strata require a larger defer region.

Discussion

This study argues that the central challenge in SA-AKI mortality modeling is not how to win a retrospective AUROC leaderboard; it is how to construct a clinically legible decision policy when the predictive ceiling is bounded. The canonical benchmark results, the grouped Optuna pass, and the added CatBoost diversity comparator all point in the same direction. Tree models remain competitive, but their gains over one another are modest and unstable enough that the real manuscript contribution comes from making uncertainty explicit rather than from claiming a dramatic architecture breakthrough.

Conformal selective triage addresses that problem directly. Instead of forcing every patient into a binary high-risk versus low-risk decision, it partitions the cohort into Alert, Clear, and Defer. That defer region is not a nuisance artifact; it is the mechanism that protects coverage. In this dataset, the defer option is exactly what allows the model to preserve reliable alert PPV and clear NPV under both clean evaluation and shift stress.

The disagreement-based comparator remains useful because it demonstrates that uncertainty-aware action restriction helps even before formal conformalization. However, the disagreement policy depends on an empirically tuned agreement threshold and is harder to justify theoretically. The conformal formulation is stronger for a journal paper because it is more principled, more general, and more transparent about what is and is not guaranteed.

From a translational standpoint, the manuscript supports a narrow but realistic deployment framing: retrospective workflow design, threshold governance, and safe automation boundaries for ICU triage. The decision-curve results support the continuous XGBoost score as the primary clinical-score comparator, while conformal and

disagreement policies should be read as downstream action-governance layers rather than as replacements for continuous ranking.

Limitations

This work has several important limitations.

- 1. The study remains internally validated only. The current checked-in cohort does not expose a defensible calendar timestamp axis for a true temporal split, and no external cohort is included.
- 2. The cohort remains affected by ETL and documentation mismatches already documented in the audit and data-contract artifacts.
- 3. Repeated patients remain present in the working cohort, which is why subject-grouped evaluation is mandatory.
- 4. The process-rich feature space may not transport as well as physiology-only inputs.
- 5. The conformal guarantees are marginal under exchangeability and do not imply perfect subgroup-conditional coverage.
- 6. Secondary horizon results, especially the 48-hour endpoint, are limited by low event counts and should be treated as sensitivity analyses rather than standalone deployment targets.
- 7. The manuscript focuses on retrospective decision support and does not evaluate clinical workflow adoption, clinician behavior change, or prospective impact.
- 8. The disagreement-based baseline is heuristic by design and is included as a comparison rather than as the final methodological recommendation.

Conclusion

In this SA-AKI cohort, the predictive ceiling for first-24-hour ICU mortality risk appears bounded, and that changes the scientific target. The most defensible contribution is not a marginal improvement in discrimination but an uncertainty-aware triage policy that states when the model should act and when it should defer. Conformal selective triage provides that framing. It yields clinically interpretable Alert/Defer/Clear decisions, finite-sample coverage guarantees, stronger reliability under conservative consensus, and more graceful degradation under shift than fixed-threshold rules. For a clinical informatics venue such as JAMIA, this is the right story: rigorous internal validation, explicit uncertainty handling, and a careful boundary around what retrospective machine learning can responsibly claim.

Data and Cohort

The audit pass identified 9002 unique subjects among 10036 working rows, with 1034 repeated-subject rows that make stay-level random splitting inappropriate for deployment claims. The package therefore treats subject-grouped evaluation as a hard design constraint rather than an optional robustness check. The explicit data contract for the journal package is intentionally conservative: one row equals one ICU-stay-level SA-AKI record scored at T24; all predictor features must be available by T24; post-T24 information is excluded from deployment claims even if it exists elsewhere in the repository; and bedside-readiness, transportability, and exact thesis-cohort equivalence are not claimed.

Table 7 Appendix Table B1. Repeated grouped benchmark summary for the main tree-family comparison.

Model	AUROC	Brier	ECE	Recall @ PPV _i =0.50
catboost	0.768	0.161	0.022	0.623
lightgbm	0.765	0.161	0.021	0.575
xgboost	0.768	0.161	0.017	0.608

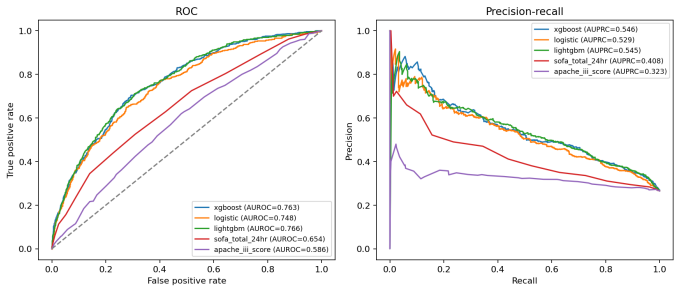


Figure 10 Appendix Figure B1. ROC and PR benchmark figure.

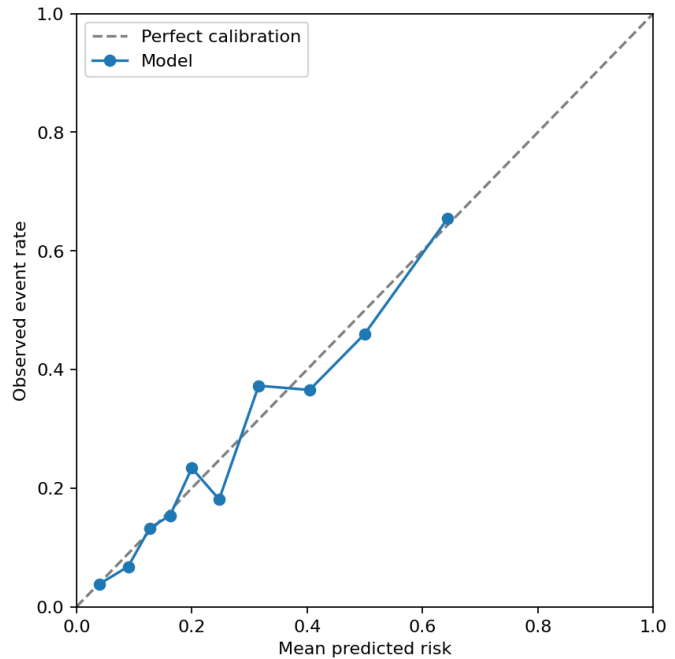


Figure 11 Appendix Figure B2. Calibration curve for the deployed baseline.

Extended Benchmarks

Appendix B gathers the benchmark tables that support the main story without overloading the primary narrative. Grouped Optuna improved the LightGBM validation operating point (validation AUROC 0.752; recall at PPV ≥ 0.50 of 0.590), but the repeated grouped benchmark still left XGBoost, tuned LightGBM, and CatBoost in a narrow performance band.

Shift and Subgroup Notes

The shift and subgroup appendices are intentionally framed as deployment notes rather than as new performance claims. Missingness

Table 8 Appendix Table B2. Grouped Optuna summary for the LightGBM benchmark.

Model	Trials	Validation AUROC	Validation recall @ PPV _l =0.50	Validation Brier	Validation ECE
lightgbm	40	0.752	0.590	0.177	0.093

Table 9 Appendix Table B3. Top grouped Optuna LightGBM trials.

Model	Trial	Objective	AUROC	Brier	Recall @ PPV _l =0.50	Trees	Learning rate	Leaves	Max depth	Min child samples
lightgbm	17	0.882	0.752	0.177	0.590	603	0.132	52	10	61
lightgbm	34	0.881	0.753	0.173	0.580	627	0.095	60	8	24
lightgbm	33	0.878	0.753	0.176	0.569	605	0.085	60	9	50
lightgbm	29	0.874	0.754	0.175	0.552	814	0.109	60	8	25
lightgbm	24	0.872	0.748	0.179	0.569	607	0.196	50	10	10

Table 10 Appendix Table B4. Feature ablation of the selected model.

feature_set	model_name	auroc	auprc	brier	ece	precision	recall	f1	specificity	alert_rate	prevalence	tp	fp	tn	fn	calibration_intercept	calibration_slope	recall_at_ppv_050	precision_at_ppv_050_threshold	threshold_ppv_050	low_risk_coverage_npv_095
physiology_severity	xgboost	0.756	0.539	0.163	0.011	0.490	0.531	0.510	0.801	0.287	0.265	282.000	293.000	1177.000	249.000	-0.058	0.986	0.552	0.500	0.365	0.173
full	xgboost	0.763	0.546	0.161	0.016	0.493	0.550	0.520	0.796	0.296	0.265	292.000	300.000	1170.000	239.000	-0.032	1.009	0.548	0.500	0.356	0.167
physiology_plus_care	xgboost	0.756	0.536	0.163	0.025	0.499	0.501	0.500	0.818	0.266	0.265	266.000	267.000	1203.000	265.000	-0.069	0.982	0.494	0.508	0.389	0.097

Table 11 Appendix Table B5. Secondary horizon sensitivity analyses.

label	prevalence	auroc	auprc	brier	ece	precision	recall	f1	specificity	alert_rate	tp	fp	tn	fn	calibration_intercept	calibration_slope	recall_at_primary_ppv
death_within_48h	0.004	0.705	0.013	0.004	0.000	0.000	0.000	0.000	1.000	0.000	0.000	0.000	2020.000	9.000	-0.168	0.964	NA
death_within_7d	0.115	0.732	0.261	0.094	0.011	0.393	0.048	0.085	0.990	0.014	11.000	17.000	1753.000	219.000	-0.160	0.891	0.060
death_before_discharge	0.265	0.763	0.546	0.161	0.016	0.493	0.550	0.520	0.796	0.296	292.000	300.000	1170.000	239.000	-0.032	1.009	0.548

Table 12 Appendix Table B6. Repeated grouped disagreement-based selective-triage summary.

Policy	Alert precision mean	Alert precision std	Alert recall mean	Alert recall std	Actionable coverage mean	Actionable coverage std	Low-risk NPV mean	Low-risk NPV std	Defer rate mean	Defer rate std	Actionable error mean	Actionable error std
fixed_threshold	0.491	0.007	0.613	0.056	0.520	0.052	0.955	0.009	0.480	0.052	0.343	0.008
selective_triage	0.484	0.025	0.225	0.151	0.260	0.139	0.960	0.013	0.740	0.139	0.254	0.033

Table 13 Appendix Table B7. Disagreement-based shift sensitivity summary.

scenario	severity	policy_name	alert_rate	low_risk_coverage	defer_rate	actionable_coverage	alert_precision	alert_recall	low_risk_npv	defer_event_rate	actionable_error_rate	high_count	low_count	defer_count
physiology_missingness_shift	0.100	fixed_threshold	0.278	0.164	0.557	0.443	0.503	0.527	0.936	0.206	0.336	557.000	329.000	1115.000
physiology_missingness_shift	0.100	selective_triage	0.093	0.128	0.779	0.221	0.543	0.190	0.926	0.264	0.235	186.000	256.000	1559.000
measurement_process_dropout	1.000	fixed_threshold	0.307	0.132	0.561	0.439	0.486	0.563	0.955	0.196	0.373	615.000	264.000	1122.000
measurement_process_dropout	1.000	selective_triage	0.100	0.100	0.800	0.200	0.475	0.179	0.955	0.267	0.285	200.000	200.000	1601.000
care_process_dropout	1.000	fixed_threshold	0.296	0.167	0.537	0.463	0.493	0.550	0.952	0.208	0.341	592.000	335.000	1074.000
care_process_dropout	1.000	selective_triage	0.106	0.133	0.761	0.239	0.514	0.205	0.951	0.269	0.243	212.000	266.000	1523.000

Table 14 Appendix Table B8. Repeated grouped clinical score summary.

model_name	auroc_mean	auroc_std	auprc_mean	auprc_std	brier_mean	brier_std	ece_mean	ece_std	recall_at_ppv_050_mean	recall_at_ppv_050_std
apache_iii_score	0.579	0.007	0.330	0.010	0.245	0.001	0.226	0.003	0.023	0.019
sofa_total_24hr	0.646	0.009	0.397	0.012	0.233	0.001	0.217	0.005	0.146	0.002

Table 15 Appendix Table B9. Clinical score operating points on the main grouped split.

score_name	auroc	auprc	brier	ece	precision	recall	f1	specificity	alert_rate	prevalence	tp	fp	tn	fn	calibration_intercept	calibration_slope	threshold_ppv_050	precision_at_ppv_050_threshold	recall_at_ppv_050_threshold	alert_rate_at_ppv_050_threshold
sofa_total_24hr	0.654	0.408	0.232	0.221	0.619	0.113	0.191	0.975	0.048	0.265	60.000	37.000	1433.000	471.000	-1.029	1.098	0.695	0.500	0.147	0.078
apache_iii_score	0.586	0.323	0.244	0.228	0.480	0.023	0.043	0.991	0.012	0.265	12.000	13.000	1457.000	519.000	-1.014	0.931	0.658	0.571	0.028	0.013

injection, care-process dropout, and measurement-process dropout all reinforce the same interpretation as the main text: conformal policies preserve coverage by deferring a larger share of difficult cases, whereas fixed thresholds lose recall without exposing uncertainty directly.

Subgroup analyses likewise reinforce a governance-oriented interpretation. Low-acuity strata support broader automation, while high-acuity strata require larger defer regions to retain reliability. These patterns do not invalidate the framework; they define where a cautious deployment policy should be most conservative.

Reproducibility

The package is generated from a fixed artifact bundle. The recommended build order is:

1. Run `saaki/deployment_analysis.py` to generate the canonical artifact bundle in `local_outputs/artifacts/`.
2. Run `saaki/build_jamia_manuscript.py` to materialize the chaptered markdown manuscript in `saaki/jamia_manuscript/`.
3. Run `saaki/build_jamia_tex.py` to copy the official OUP files, regenerate the figure/table assets, and write `main.tex`, `refs.bib`, and the package README.

Table 16 Appendix Table B10. Key decision-curve net-benefit comparisons.

Threshold	Strategy	Net benefit	Standardized net benefit	Alert rate
0.100	apache_iii_score_continuous	0.184	0.692	1.000
0.100	conformal_alpha_0.05	0.064	0.242	0.107
0.100	disagreement_selective	0.045	0.170	0.099
0.100	fixed_threshold_ppv50	0.129	0.487	0.296
0.100	sofa_total_24hr_continuous	0.184	0.692	1.000
0.100	union_conformal_alpha_0.05	0.037	0.138	0.050
0.100	xgboost_continuous	0.194	0.730	0.821
0.200	apache_iii_score_continuous	0.082	0.308	1.000
0.200	conformal_alpha_0.05	0.059	0.222	0.107
0.200	disagreement_selective	0.038	0.145	0.099
0.200	fixed_threshold_ppv50	0.108	0.409	0.296
0.200	sofa_total_24hr_continuous	0.082	0.308	1.000
0.200	union_conformal_alpha_0.05	0.035	0.131	0.050
0.200	xgboost_continuous	0.133	0.500	0.549
0.300	apache_iii_score_continuous	-0.049	-0.184	0.997
0.300	conformal_alpha_0.05	0.052	0.196	0.107
0.300	disagreement_selective	0.030	0.112	0.099
0.300	fixed_threshold_ppv50	0.082	0.308	0.296
0.300	sofa_total_24hr_continuous	-0.034	-0.130	0.952
0.300	union_conformal_alpha_0.05	0.033	0.123	0.050
0.300	xgboost_continuous	0.092	0.345	0.364

Table 17 Appendix Table B11. Fixed-policy clinical utility summary.

Strategy	Alert rate	Alert precision	Alert recall	Net benefit @0.10	Net benefit @0.20	Net benefit @0.30
treat_all	1.000	0.265	1.000	0.184	0.082	-0.049
treat_none	0.000	NA	0.000	0.000	0.000	0.000
fixed_threshold_ppv50	0.296	0.493	0.550	0.129	0.108	0.082
disagreement_selective	0.099	0.510	0.190	0.045	0.038	0.030
conformal_alpha_0.05	0.107	0.640	0.258	0.064	0.059	0.052
conformal_alpha_0.10	0.157	0.589	0.348	0.085	0.076	0.065
union_conformal_alpha_0.05	0.050	0.752	0.143	0.037	0.035	0.033

Table 18 Appendix Table B12. Negative and null probe results carried forward for transparency.

Probe	AUROC	AUPRC	Recall @ PPV _i =0.50
feat_missingness	0.755	0.537	0.563
feat_ratios	0.751	0.526	0.514
ensemble_stack_3	0.759	0.537	0.544
hpo_lgbm	0.760	0.540	0.595
survival_rsf	0.726	NA	NA

The selected model snapshot preserved in the artifact bundle is XGBoost with isotonic calibration, mean AUROC 0.768, mean AUPRC 0.544, mean recall at PPV ≥ 0.50 of 0.608, and decision-curve net benefit 0.133 at threshold 0.20. The source patient-level data are not redistributed with this repository; reproducing the cohort from raw source requires independent credentialed access to MIMIC and the corresponding data-use approvals.

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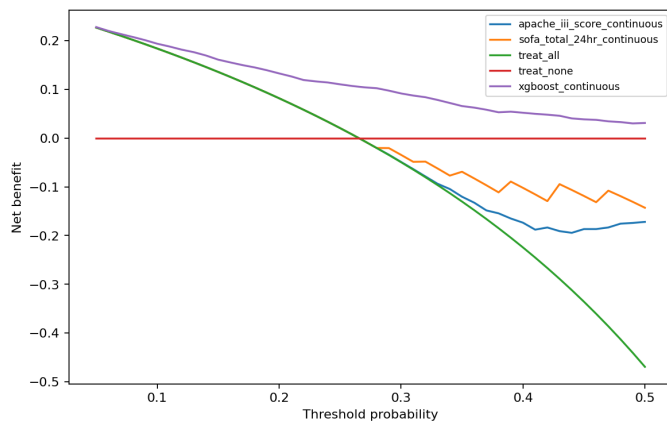


Figure 12 Appendix Figure B3. Decision curve for benchmark and fixed-policy comparators.

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